Muhammad Asif Jamal



Medical Image Analysis and Computer Vision Researcher

Chosun University, Gwangju, South Korea

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Profile

PhD researcher in Information and Communication Engineering, expecting to graduate in February 2027 and seeking a postdoctoral position in medical image analysis and computer vision. Research experience includes in 3D dental image segmentation, attention mechanisms, state space models and 3D reconstruction. Research focuses on tooth localization and instance segmentation in cone beam computed tomography (CBCT). Previous university teaching experience includes undergraduate and graduate instruction, project supervision, and development of a high performance computing laboratory.

Areas of Expertise

Medical Image Analysis • Computer Vision • Deep Learning 3D Dental Segmentation • Attention Mechanisms • State Space Models

Research Experience

Graduate Research Associate — *Mar 2023 to present*
ENVI Lab, Chosun | KENTECH — South Korea

- Develop deep learning methods for tooth localization and instance segmentation in 3D dental CBCT volumes under the supervision of Prof. Bumshik Lee.
- Design DentSegPlus, a staged framework combining region localization, centroid and skeleton prediction, and individual tooth segmentation using enhanced dual attention and axis-wise Mamba blocks.

Research Collaborator (Remote) — *Aug 2025 to present*
Intelligent Media Center, Air University — Islamabad, Pakistan

- Collaborate remotely with students on computer vision and machine learning research, providing research guidance through online meetings.

Professional Experience

Lecturer — *Sep 2016 to Feb 2026*
Air University, Islamabad — Pakistan

- Taught undergraduate and graduate students in OOP, data structure and algorithms, theory of automata, assembly language, cloud computing, scientific writing, and digital logic design.
- Supervised student theses and undergraduate projects.

Lecturer and Project Coordinator — *Aug 2015 to Aug 2016*
Abasyn University, Peshawar — Pakistan

- Taught undergraduate courses and supervised student projects.
- Coordinated the establishment of a High Performance Computing (HPC) Laboratory comprising six compute nodes, one head node, and one domain controller. Led a five-person team in developing the computing cluster lab.

Database Administrator, part time — *Feb 2011 to May 2015*
Tidningstjänst AB, Sundsvall — Sweden

Managed worker records, duty scheduling, and customer subscription data.

Education

PhD in Information and Communication Engineering *Mar 2023 to Feb 2027 (expected)*
Chosun University, Gwangju South Korea
CGPA: 4.33/4.50.
Supervisor: Prof. Bumshik Lee.
Research: Anatomy-Guided and Context-Adaptive Representation Learning for Robust 3D Tooth Instance Analysis in Cone-Beam CT.

Master's in Computer Engineering *Sep 2010 to Aug 2013*
Mid Sweden University, Sundsvall Sweden
Thesis: A TinyOS Testbed for CC2420 Transceiver Chip Using NesC Programming. Full Thesis
Activities: Member of the MIUN Student Union.

Bachelor's in Computer Systems Engineering *Aug 2005 to Jul 2009*
University of Engineering and Technology, Peshawar Pakistan
Final Project: Design and Simulation of OFDM Signalling Using LabVIEW.

Publications

Journal Articles

- Maha Abdelhaq, Aysha Naseer, Raed Alsaqour, Ahmad Jalal, **Muhammad Asif Jamal**, Aamir Nadeem, and Jinsul Kim. "A novel deep learning-based object detection along semantic segmentation on aerial imagery." *PeerJ Computer Science*, 12, e3648, 2026. DOI: 10.7717/peerj-cs.3648.
- **Muhammad Asif Jamal**, Guoqing Chao, and Bumshik Lee. "DentSeg-EDA: A 3D tooth segmentation framework for CBCT images using enhanced dual attention." *Neurocomputing*, 655, 131437, 2025. DOI: 10.1016/j.neucom.2025.131437.
- Asifa Mehmood Qureshi, Moneerah Alotaibi, Sultan Refa Alotaibi, Dina Abdulaziz AlHammadi, **Muhammad Asif Jamal**, Ahmad Jalal, and Bumshik Lee. "Autonomous vehicle surveillance through fuzzy C-means segmentation and DeepSORT on aerial images." *PeerJ Computer Science*, 11, e2835, 2025. DOI: 10.7717/peerj-cs.2835.
- Mouazma Batool, Moneerah Alotaibi, Sultan Refa Alotaibi, Dina Abdulaziz AlHammadi, **Muhammad Asif Jamal**, Ahmad Jalal, and Bumshik Lee. "Multimodal Human Action Recognition Framework Using an Improved CNNGRU Classifier." *IEEE Access*, 12, pp. 158388 to 158406, 2024. DOI: 10.1109/ACCESS.2024.3481631.
- Syed Atif Moqurrab, Tariq Naeem, M. Shoaib Malik, Asim Ali Fayyaz, **Asif Jamal**, and Gautam Srivastava. "UtilityAware: A framework for data privacy protection in e-health." *Information Sciences*, 643, 119247, 2023. DOI: 10.1016/j.ins.2023.119247.
- **M. Asif Jamal**, Aziz Ur Rehman, Moonisa Ahsan, M. S. Riaz, and M. S. Zafar. "Initialization Method for Communication and Data Sharing in P2P Environment Between Wireless Sensor Nodes." *International Journal of Advanced Computer Science and Applications*, 9(8), 2018. DOI: 10.14569/IJACSA.2018.090880.

Conference Papers

- **Muhammad Asif Jamal** and Bumshik Lee. "Accurate 3D Tooth Segmentation via Skeleton and Centroid Guidance in CBCT Imaging." In *Proceedings of the 7th International Conference on Artificial Intelligence in Information and Communication (ICAIIC)*, pp. 567 to 571, Fukuoka, Japan, 2025. DOI: 10.1109/ICAIIC64266.2025.10920685.
- Syed Sarmad Ali, **Muhammad Asif Jamal**, Muhammad Yaseen Khan, Syed Muhammad Asim Ali Rizvi, and Ziyu Wang. "Software Estimation Process Improvement using Data Segmentation." In *Proceedings of the 20th International Bhurban Conference on Applied Sciences and Technology (IBCAST)*, IEEE, pp. 279 to 284, 2023. Paper.
- **Muhammad Asif Jamal** and Bumshik Lee. "MambaTeeth: 3D Dental Segmentation using State Space Models with SE Attention." In *Proceedings of the Korean Institute of Next Generation Computing Spring Conference*, pp. 51 to 52, May 2025. Paper.
- **Muhammad Asif Jamal** and Bumshik Lee. "Tooth Region Extraction from CBCT Images Using 3D Binary Segmentation." In *Proceedings of the Korean Institute of Communications and Information Sciences (KICS) Summer Conference*, pp. 986 to 987, June 2024. Paper.

Manuscripts in Submission Process

- **Muhammad Asif Jamal** and Bumshik Lee. "DentSegPlus: Centroid and Skeleton-Guided Tooth Segmentation with Cross-Axis Context Modeling." Manuscript in preparation for submission.
- Fawad, **Muhammad Asif Jamal**, and Bumshik Lee. "Projection-Guided Latent Diffusion and Refinement for Sparse-View 3D CT Synthesis." Manuscript in preparation for submission.

Selected Research Projects

- **DentSegPlus:** Centroid and Skeleton-Guided Tooth Segmentation with Cross-Axis Context Modeling. Developed a pipeline for tooth localization and individual tooth segmentation in CBCT.
- **Dental reconstruction research:** Investigate reconstruction of 3D dental structures from panoramic radiographs and sparse or low dose CT inputs.
- **Breast Cancer Segmentation Using Two-and-a-Half U-Net, 2024:** Led Team Segmed in developing a hybrid segmentation architecture that extends a modified Double U-Net with a Half U-Net for feature refinement. Evaluated the model using 150 training samples and 50 testing samples, achieving a global Dice score of 67.19% and mean IoU of 57.75%. The team won the K-Health MAITHON competition and received **KRW 3 million**.
- **National Center for Cyber Security Establishment Proposal, 2017:** Prepared and submitted a proposal to the Higher Education Commission (HEC) for establishing the National Center for Cyber Security at Air University, under the supervision of Prof. Dr. Kashif Kifayat, the Dean and Director NCCS. The proposal resulted in approval for establishing the center at Air University.
- **High Performance Computing Laboratory, 2015:** Planned and implemented a university computing cluster with a five-person team.

Achievements

- **ROOKIE Research Support Grant, Chosun University, 2026:** Awarded **KRW 6 million** to support doctoral research activities and international journal publication expenses.
- **Winner, K-Health MAITHON 2024:** Led the winning team in the breast cancer segmentation competition, receiving a prize of **KRW 3 million**.
- **Full Doctoral Scholarship, Chosun University:** Awarded full PhD funding covering tuition fees, a living stipend, and health insurance.
- **Faculty Development Program (FDP) Scholarship, Air University:** Awarded a scholarship to pursue PhD studies.
- **Best Teacher Award, Air University, 2022:** Recognized by the Department of Computer Science, Faculty of Artificial Intelligence.

Conference, Seminar and Workshops Participation

- **KDD 2026:** Attended the main KDD Conference on knowledge discovery and data mining, data science, machine learning, and artificial intelligence, Jeju, South Korea.
- **MICCAI 2025:** Attended the main International Conference on Medical Image Computing and Computer Assisted Intervention, Daejeon, South Korea.
- Attended a one-day Google Startup boot camp for web developers in Beijing, China.
- Delivered a seminar titled "How to Get Admission in Europe" at Abasyn University, Peshawar, to an audience of more than 60 students and teachers.
- Attended the "FPGA Based System Design" workshop offered by the Military College of Signals, NUST, Islamabad.

Skills

- **Research:** Medical image segmentation, dataset preparation, experimental comparison, ablation studies, and academic writing.

- **Programming and Deep Learning:** Python, PyTorch, C, C++, Java, MATLAB, NesC, and embedded C.
- **Tools and Systems:** Linux, LabVIEW, MySQL, HTML, CSS, PHP, and WordPress.

Hobbies and Interests

Inline skating, skateboarding, table tennis, reading, traveling, and camping.

Languages

Urdu: Native. English: Professional proficiency. Korean: Studying KIIP Level 2. Swedish: Basic conversational. Arabic: Basic conversational.

References

Available upon request.